PRATHAMESH SHINDE

Computer Science Student | AI/ML Engineer | Computer Vision | LLMs & Agentic AI | Full Stack Developer
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SUMMARY

Third-year Computer Science Engineering student at Sanjivani University. I enjoy building AI that people can actually use, mainly in Computer Vision and Large Language Models.

I worked full-time for five months at Ethara AI, rating AI-generated images, videos and audio and writing prompts to help improve AI models. I also led a six-member team on KrushiAI, a crop disease detection system, and I am the first author of our research paper on it, accepted at IC2NC 2026.

I have built and deployed full-stack AI web apps with React, TypeScript, Node.js and FastAPI using the Gemini and Claude APIs, and designed offline edge-AI systems for national hackathons. I want to grow as a researcher in a strong lab and work on real research problems.

ACHIEVEMENTS

• First-author research paper **accepted at IC2NC 2026** on AI-based crop disease detection
• Built **5+ AI and full-stack projects**, 3 of them live: KrushiAI, SkillMap AI and Paper Trail
• Completed a 5-month full-time internship at Ethara AI, working on human feedback data for AI models
• Led a 6-member team on KrushiAI, taking it from an AI model to a live app and an accepted research paper
• Filed a copyright registration with the Indian Copyright Office (2026, under review)
• 2nd place in a university-level web development hackathon
• Took part in 5+ national hackathons, including Smart India Hackathon 2026, Tata Technologies InnoVent-27 and the OpenAI × Outskill AI Builders Hackathon

EDUCATION

Bachelor of Technology – Computer Science and Engineering
Sanjivani University, Kopargaon (2024 – 2028)
CGPA: 8.03 / 10
Higher Secondary Certificate (Class XII)
Vidya Vikas Secondary College (2023 – 2024)
Percentage: 75%

SKILLS

Programming: Python, Java, C++, SQL, JavaScript, TypeScript, HTML
AI / ML: Deep Learning, Machine Learning, Computer Vision, CNN, Transfer Learning (MobileNetV2), Image Processing, TensorFlow/Keras, PyTorch, Model Evaluation
LLMs & Agentic AI: LangGraph, RAG, Multi-Agent Systems, ChromaDB, Gemini API, OpenAI API, Claude API, Prompt Engineering
Full Stack: React, Angular, Node.js, Express, FastAPI, Streamlit, REST APIs
Databases: MongoDB Atlas, PostgreSQL, MySQL
Edge AI: Raspberry Pi, NVIDIA Jetson Nano, ESP32
Web Design: Responsive Layouts, Mobile-Friendly Interfaces, User-Centered Design, Clean Dashboards
Automation: n8n Workflows, UiPath (RPA), Python Scripts for Repetitive Tasks
Tools: Git, GitHub, LaTeX, Vercel, Render

PROJECTS

1) KrushiAI – AI Crop Disease Detection Platform

Live URL: krushiai.streamlit.app
Tech Stack: Python, TensorFlow/Keras, MobileNetV2, Streamlit, MongoDB Atlas

- Led a team of 6 to build an app that spots pomegranate diseases from a single photo.
- Trained the model on 5,099 images from a public Kaggle dataset, covering 4 diseases and healthy fruit.
- Rebuilt our first model (about 500 MB) with MobileNetV2 transfer learning, making it about 50× smaller (9.59 MB).
- Reached 93.92% validation accuracy and a 0.93 macro F1-score, using early stopping to avoid overfitting.
- Added login, scan history, weather, crop advice and government scheme pages in more than one language, with data stored in MongoDB Atlas.

2) SkillMap AI – AI Resume & ATS Analyzer

Live URL: skillmapai-yu8e.onrender.com
Tech Stack: React, TypeScript, Node.js, Express, PostgreSQL (Prisma), Google Gemini API

- Built a web app where you upload your resume and paste a job description, and it gives you an ATS match score, the skills you have and the ones you're missing.
- Used Gemini with a fixed JSON response format so results come back clean and consistent, with a backup model when the main one is busy or out of quota.
- Added sign-up and login with secure sessions, password reset by email, and a dashboard to track your scores over time.
- Suggests free courses and tips for each missing skill, and deployed it on Render with a Neon PostgreSQL database.

3) Paper Trail – Multilingual Medical Report Explainer

Live URL: paper-trail-dusky.vercel.app
Tech Stack: React, TypeScript, Node.js (Vercel serverless), Google Gemini API, MongoDB Atlas, Clerk

- Built a web app that reads a blood report PDF and explains each result in simple Marathi, Hindi or English.
- Used the Gemini API to pull out the values, explain medical terms in plain words, and flag results that need a doctor's attention urgently.
- Added read-aloud using the device's built-in voice, so people who find reading hard can listen instead.
- Added charts that show how results change across reports, plus sign-in to keep each person's reports private. PDFs are deleted right after processing.
- First built at the OpenAI × Outskill AI Builders Hackathon, then rebuilt to run fully on free services.

4) SevaPath – Government Jobs & Exam Preparation Platform

GitHub: github.com/PrathameshShinde24/SevaPath
Tech Stack: React, FastAPI, MongoDB, Claude API

- Built a platform that helps students in Maharashtra find government jobs and prepare for competitive exams.
- Made the full app myself: React for the frontend, FastAPI for the backend and MongoDB for the database.
- Added AI-based guidance using the Claude API to help students with their questions.
- Added Marathi support so students can use the platform in their own language.
- Now testing with sample data while exam rules are checked; public launch is the next step.

5) SafarSathi & TyreSense – Edge AI System Designs (Starting Phase)

Tech Stack: Raspberry Pi, Jetson Nano, ESP32, OBD-II, MEMS Accelerometers

- SafarSathi: designed an offline car diagnosis assistant on Raspberry Pi / Jetson Nano that reads OBD-II data.
- SafarSathi explains faults by voice in Hindi and Marathi and works without internet.
- TyreSense: designed a tyre health monitor that uses an ESP32 and MEMS motion sensors to check tyres on the device itself.
- Prepared full system designs and pitch decks for both at the Tata Technologies InnoVent-27 hackathon.

RESEARCH & PUBLICATIONS

Research Paper — KrushiAI, IC2NC 2026 ***(Accepted)***

- First author, with Akash Pingale, under the guidance of Prof. Patolla Venkat Reddy.
- Title: A Deep Learning Approach for Early Detection and Classification of Pomegranate Fruit Diseases Using Convolutional Neural Networks.
- Lightweight MobileNetV2 model: 93.92% validation accuracy, 0.93 macro F1-score, 9.59 MB model size.

Research Proposal — Cipher-Encoded Jailbreak Resistance in LLMs ***(In Progress)***

- Studying how AI chat systems can catch harmful requests hidden inside codes or ciphers, without blocking normal translation, encoding and puzzle tasks.
- Builds on Yuan et al., “GPT-4 Is Too Smart to Be Safe” (ICLR 2024).

INTERNSHIP EXPERIENCE

LLM Post-Training Intern — Ethara AI (Full-time) ***(Feb 2026 – Jun 2026)***

- Worked full-time (8 hours/day) for 5 months on human feedback data for multimodal AI models.
- Compared and rated AI-generated images, videos and audio every day, following clear written guidelines.
- Wrote prompts and detailed image descriptions used as feedback to improve AI models.
- Learned how much careful human judgment and clean data matter in making AI models better.

CERTIFICATIONS

- NPTEL – The Joy of Computing using Python (with Distinction)
- Google Cloud – Exploring Data Transformation with Google Cloud, Google ML Crash Course
- Microsoft – Azure AI Engineer Associate, Introduction to GenAI
- MongoDB – Introduction to MongoDB for Students
- Simplilearn – Introduction to C++, DevOps, Java Programming

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